Inteligência Computacional:
Naïve Bayes and Bayesian Learning

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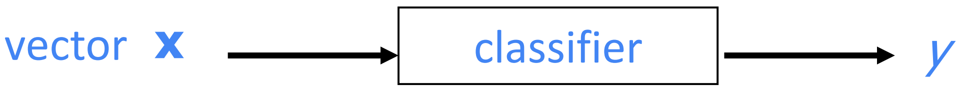
Nov. 23, 2020

Classification

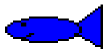
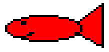
Problem:

- Given a vector **x** with *N* parameters, find its class *y* among *Y* possible labels

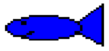
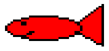
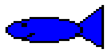


or

Fish classifier:

 or  ?

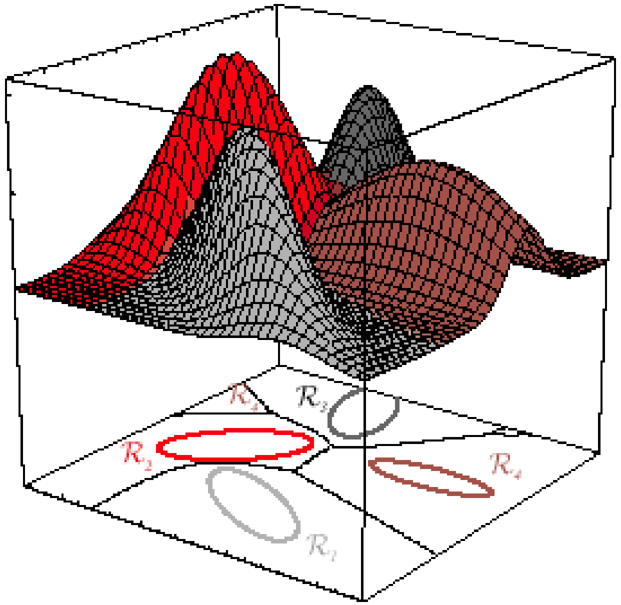
Training set used during ➔ the training stage

Length	Weight	Class *y*
12	3.2	
10	0.5	
14	2.8	

Test stage uses “unseen” examples:
$\boldsymbol{x}$ = (13, 4.2)
y = ?

Evaluation: misclassification (or error) rate

Statistical Bayes classifiers



- Based on Bayes' theorem, requires the estimation of likelihoods and priors for each class:

What we want to know; the Posterior probability of Class j given features x

The Likelihood; the probability of the features given a Class j.

The Prior probability of Class j; what we know about the class distribution before we consider x.

$$P(Class_j \mid x) = \frac{P(x \mid Class_j) \times P(Class_j)}{P(x)}$$

The Evidence. In practice, there's interest only in the numerator (denominator is effectively constant)

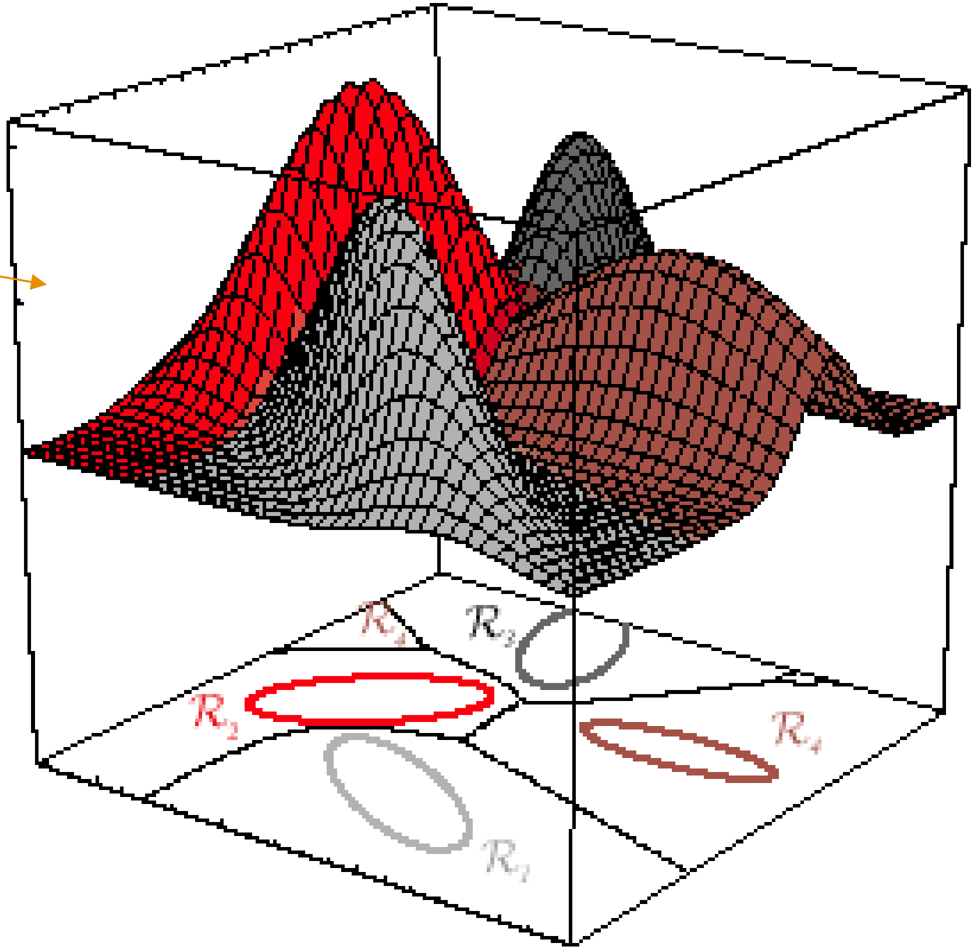
- If the estimation of likelihoods and priors is perfect, the Bayes classifier is optimum and its error is the minimum possible, called "Bayes error"

http://shatterline.com/blog/2013/09/12/not-so-naive-classification-with-the-naive-bayes-classifier/

Example of a Bayes classifier

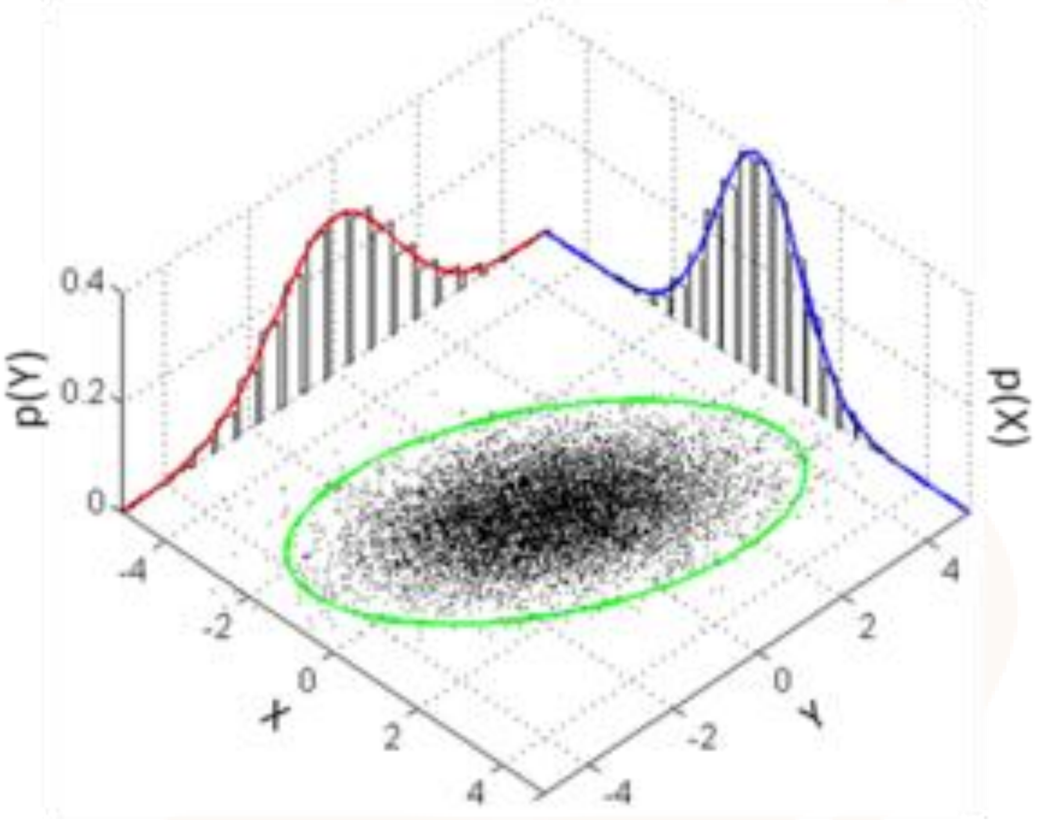
- Example: 4 classes
- "Nature" dictated that in this case each likelihood is a Gaussian distribution
- Assume the priors are uniformly distributed $P(Class_j)$=1/4
- Given a test vector x, calculate posteriors for j=1, 2, 3, 4 and choose the largest:

$$P(Class_j \mid x) = \frac{P(x \mid Class_j) \times P(Class_j)}{P(x)}$$



Naïve Bayes classifier

- Use the naïve (ingenuous) assumption that the parameters of each class are statistically independent, such that all **joint** distributions are equivalent to the multiplication of the **marginal** distributions
  - Example, for a parameter vector **z** = [x, y] of a classe *C*, it corresponds to
    $$P(\mathbf{z}|C) = P(x|C)\,P(y|C)$$



Simple classifiers (for two simple sets)

- Instead of using an automatic training method, let us design simple classifiers ourselves using the two simple sets below:

Training set

Length	Weight	Class y
12	3.2	0
10	0.5	1
14	2.8	0
14	2.4	0
13	1.8	1
13.8	1.5	0
11	1	1

Test set

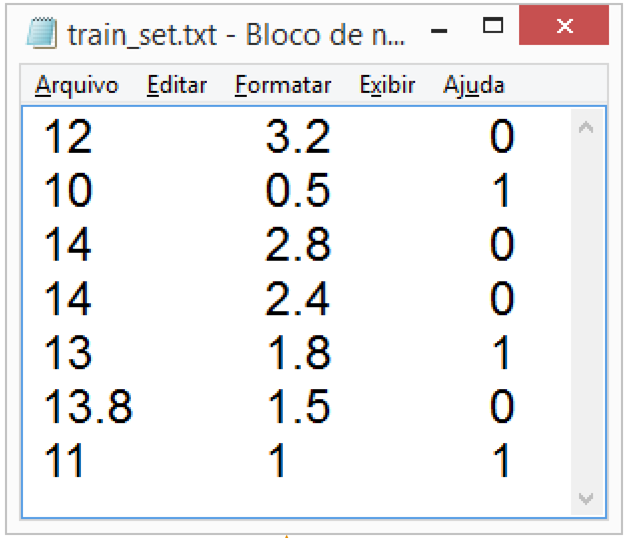
Length	Weight	Class y
13	3.1	0
9	0.8	0
12.3	1.4	1
10	2.3	1

Getting familiar with the given data

- Because there are (only) two features, it is easy to visualize the training set

Training set

Length	Weight	Class y
12	3.2	0
10	0.5	1
14	2.8	0
14	2.4	0
13	1.8	1
13.8	1.5	0
11	1	1


train_set.txt - Bloco de n...
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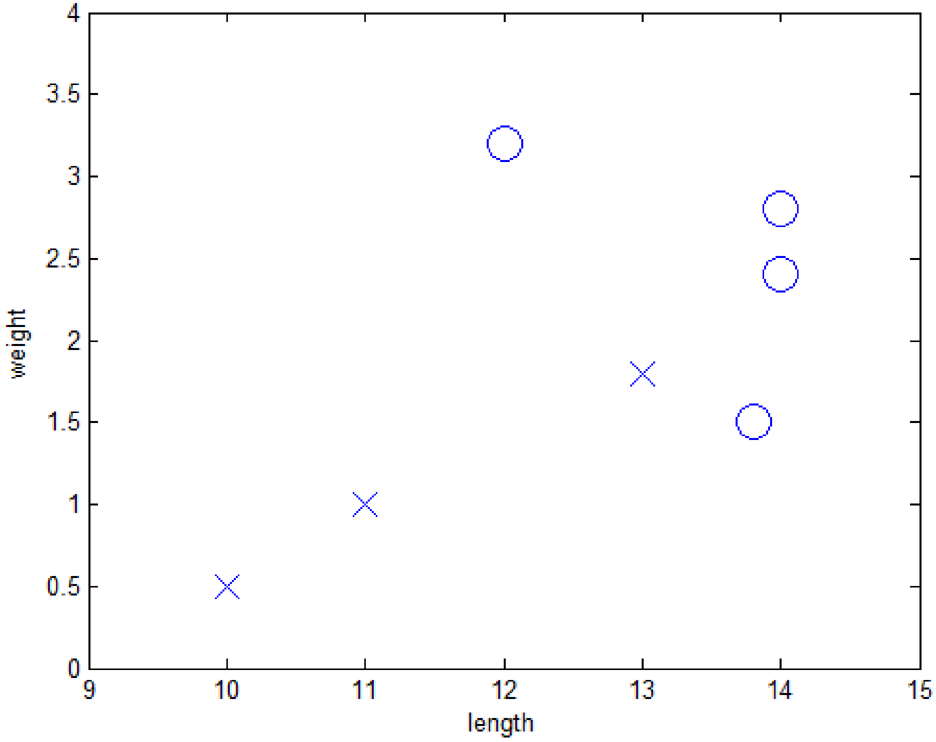
12	3.2	0
10	0.5	1
14	2.8	0
14	2.4	0
13	1.8	1
13.8	1.5	0
11	1	1

Text (ASCII) file



Manipulating the data

- Matlab / Octave code showSets.m to plot the data:

x is a 7x3 array

```
x=load('train_set.txt'); %load from file to memory
%x=load('test_set.txt'); %load from file to memory
indices0=find(x(:,3)==0); %find instances with class label == 0
indices1=find(x(:,3)==1); %find instances with class label == 1
x0=x(indices0,1:2); %label 0: ony two features, without last column
x1=x(indices1,1:2); %label 1: ony two features, without last column
clf %clear figure
plot(x0(:,1),x0(:,2),'o','MarkerSize',14) %plot label 0
hold on %superimpose plots
plot(x1(:,1),x1(:,2),'x','MarkerSize',14) %plot label 1
axis([9 15 0 4]) %resize the x and y axes
xlabel('length'), ylabel('weight')
```

train_set.txt - Bloco de n...

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12	3.2	0
10	0.5	1
14	2.8	0
14	2.4	0
13	1.8	1
13.8	1.5	0
11	1	1

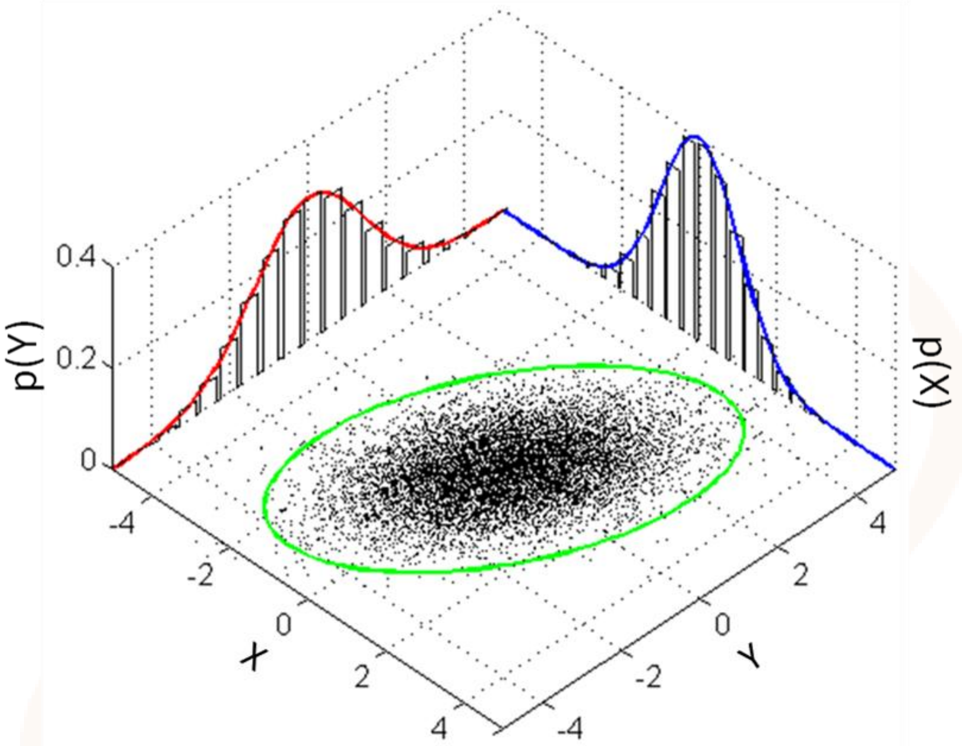
Naïve classifier using Gaussians for real-valued features

- For each class, estimate the needed statistics, priors and likelihoods
- Class 0:
  - prior0=4/7
  - m0l=mean([12 14 14 13.8]) %13.450
  - m0w=mean([3.2 2.8 2.4 1.5]) %2.4750
  - s0l=std([12 14 14 13.8]) %0.97125
  - s0w=std([3.2 2.8 2.4 1.5]) %0.72744
- Class 1:
  - prior1=3/7
  - m1l=mean([10 13 11]) %11.333
  - m1w=mean([0.5 1.8 1]) %1.1000
  - s1l=std([10 13 11]) %1.5275
  - s1w=std([0.5 1.8 1]) %0.65574
- Test example: x=[13 3.2]; xl=x(1); xw=x(2);
  - num0=normpdf(xl,m0l,s0l)*normpdf(xw,m0w,s0w)*prior0
  - num1=normpdf(xl,m1l,s1l)*normpdf(xw,m1w,s1w)*prior1
  - Px=num0+num1; %denominator, just a scale factor
  - posterior0=num0/Px %0.99685
  - posterior1=num1/Px %0.0031541

train_set.txt - Bloco de n...

Arquivo Editar Formatar Exibir Ajuda

12	3.2	0
10	0.5	1
14	2.8	0
14	2.4	0
13	1.8	1
13.8	1.5	0
11	1	1

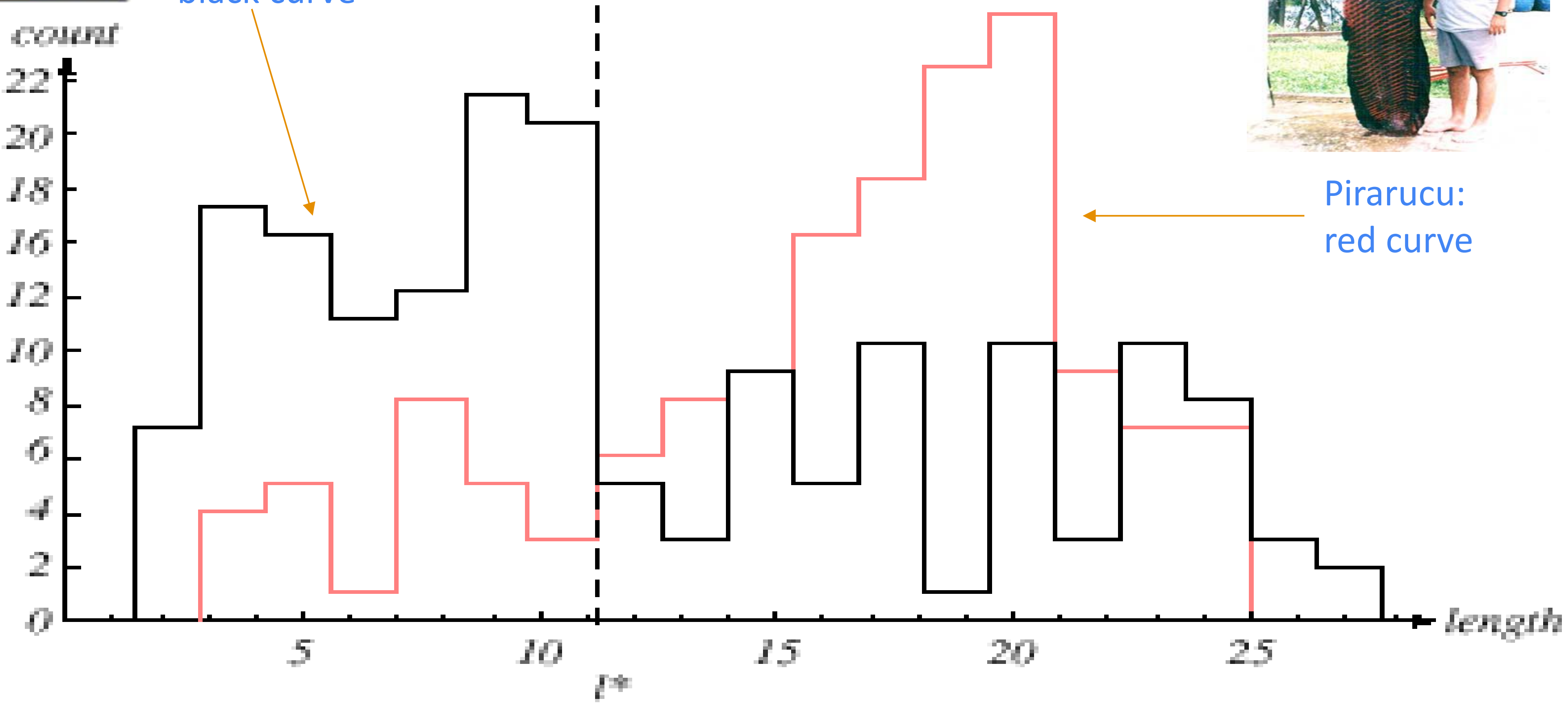


Classification based on histograms for discrete features

Piranha:
black curve

Pirarucu:
red curve

Histogram of the discretized "length" feature



Thanks!

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